Arsalene Zbidi

AI ENGINEER · PhD RESEARCHER
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PROFILE

AI Engineer and PhD researcher building agentic systems, applied machine learning, and production-oriented backend software. Experience spans ERP-integrated assistants, document-grounded retrieval, computer vision, and AI solution architecture. Research focus: agentic AI for cybersecurity threat hunting.

EXPERIENCE

Head of AI | **Mar 2026 - Present**
Microzed Solution | Megrine, Tunisia

- Lead a team of two AI engineers building agents integrated with the Zsolution ERP platform.
- Guide development of an ERP assistant for cross-module analytics, conversational queries, and system workflows.

AI Engineer & Solution Architect | **Oct 2024 - Mar 2026**
Artificial Insight | Tunis, Tunisia

- Led technical strategy for a five-person engineering team working on the PowerSale AI platform.
- Developed Telco GPT conversational AI with LangGraph multi-agent orchestration.
- Built a multi-tenant retrieval system for Kuwait Fire Force training material, combining Qdrant semantic retrieval, Elasticsearch keyword search, Arabic OCR, and graph memory.
- Developed a multilingual meeting pipeline for speech transcription, summaries, and text-to-speech reports.

Machine Learning Engineer Intern | **Feb 2024 - Sep 2024**
Artificial Insight | Tunis, Tunisia

- Built vehicle recognition and license-plate OCR workflows with YOLOv8, OpenCV, EasyOCR, and TensorFlow.
- Integrated SORT tracking with MongoDB alerting for vehicle identification.

DevOps Engineer Intern | **Jan 2021 - Jun 2021**
Polina Group Holding | Tunis, Tunisia

- Worked on CI/CD and monitoring using GitLab CI, Jenkins, ArgoCD, Kubernetes, Grafana, Prometheus, and SonarQube.

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SELECTED PROJECTS

MICRO - ERP analytics assistant | LangGraph, MCP, FastAPI, MySQL
Agent architecture connects a LangGraph assistant to ERP operations through a separate FastMCP server; employee and AI permissions are checked before tool execution.

FootTraffic AI | Python, ONNX Runtime, OpenCV, FastAPI
Built a doorway counting pipeline using YOLO ONNX detection, ByteTrack, and a hysteresis band, with a multi-camera API and event reporting. A documented 13-crossing calibration sample reached 13/13 at the selected band width.

Research Pal | Python, PDF retrieval, LLMs
Developed PDF-grounded reading, search, and chat workflows that verify generated quotations against extracted document text and flag unsupported claims.

ERP order dispatching and sourcing | FastAPI, Gemini, MySQL
Built ERP APIs for stock-aware order dispatching, warehouse sourcing, and product descriptions; stock calculations are prepared server-side before AI recommendations.

EDUCATION

PhD in Computer Science - Cybersecurity & AI | 2024 - Present
Higher School of Communications of Tunis (SUP'COM) | Research: threat hunting using agentic AI

Engineering Degree in Computer Science - Data Science & AI | 2021 - 2024
Central Polytechnic School of Tunis

Bachelor in Computer Science | 2018 - 2021
Higher Institute of Information and Communication Technologies

TECHNICAL SKILLS

AI / ML: LangGraph, LangChain, RAG, multi-agent systems, MCP, NLP, YOLOv8, TensorFlow, PyTorch, OpenCV
Backend / data: Python, Java, FastAPI, PostgreSQL, MySQL, MongoDB, Elasticsearch, Redis, Qdrant, FalkorDB
Infrastructure: Docker, Kubernetes, GitLab CI, Jenkins, ArgoCD, Prometheus, Grafana, SonarQube
Languages: Arabic (native), English (fluent), French (fluent)

LEADERSHIP

Captain, Tunisia National Volleyball Team (2014-2018); captain, Hammam-lif Senior Volleyball Team (2016-present).